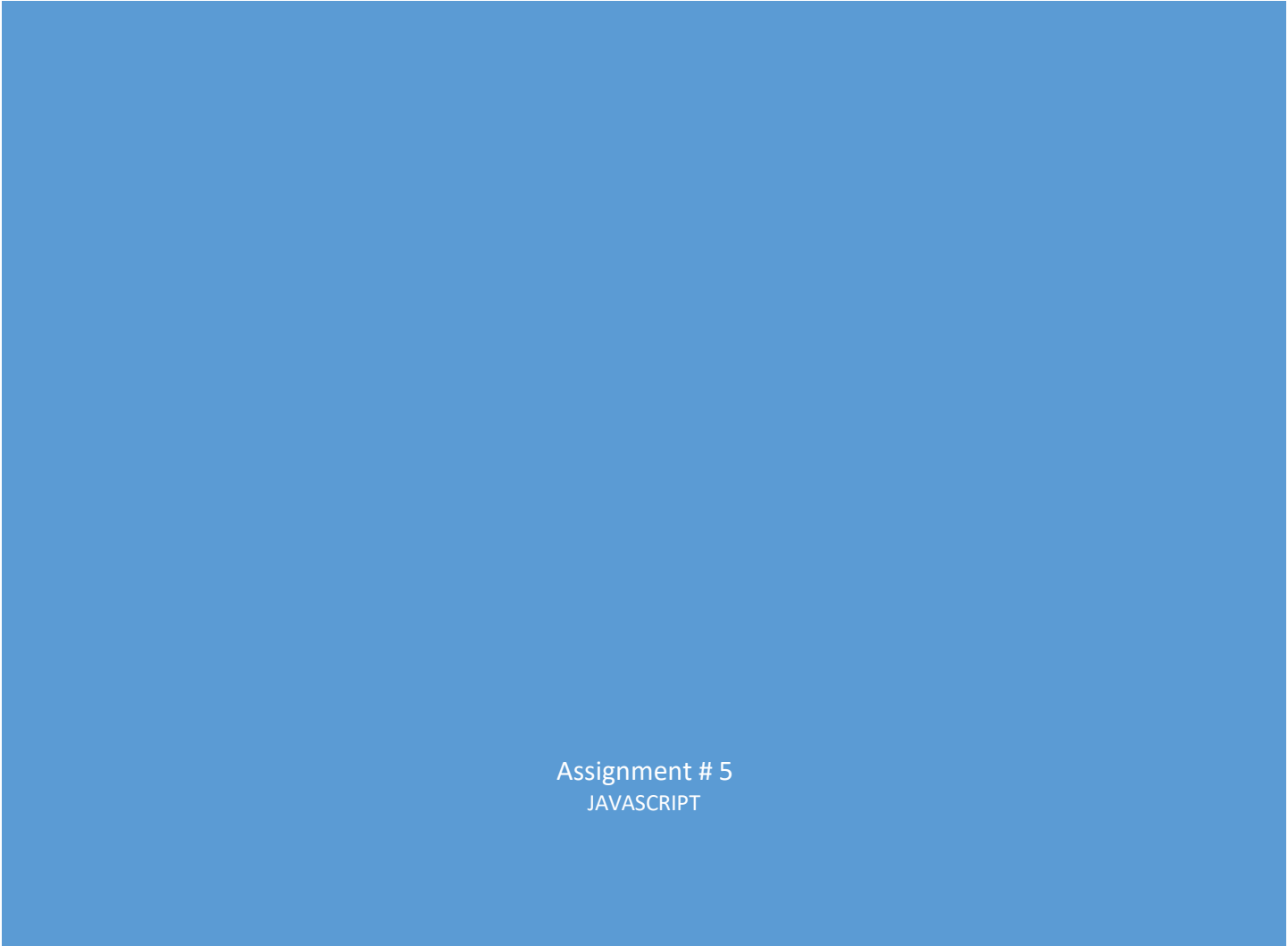




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Assignment # 5
JAVASCRIPT

1. Write a program that take two numbers & add them in a new variable. Show the result in your browser.



Sum of 3 and 5 is 8

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 1</title>
</head>
<body>

<h2>Addition of Two Numbers</h2>

<script>
let num1 = 15;
let num2 = 5;
let sum = num1 + num2;

document.write("First Number: " + num1 + "<br>");
document.write("Second Number: " + num2 + "<br>");
document.write("Sum: " + sum);
</script>

</body>
</html>
```

Output:

Addition of Two Numbers

First Number: 15
Second Number: 5
Sum: 20

2. Repeat task1 for subtraction, multiplication, division & modulus.

Code:

```
<script>

let num1 = 15;
let num2 = 5;

document.write("First Number: " + num1 + "<br>");
document.write("Second Number: " + num2 + "<br><br>");

document.write("Addition = " + (num1 + num2) + "<br>");
document.write("Subtraction = " + (num1 - num2) + "<br>");
document.write("Multiplication = " + (num1 * num2) + "<br>");
document.write("Division = " + (num1 / num2) + "<br>");
document.write("Modulus = " + (num1 % num2));

</script>

</body>
</html>
```

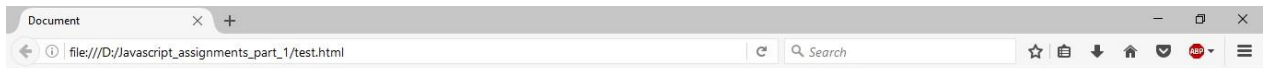
Output:

Arithmetic Operations

First Number: 15
Second Number: 5

Addition = 20
Subtraction = 10
Multiplication = 75
Division = 3
Modulus = 0

3. Do the following using JS Mathematic Expressions Declare a variable.



Value after variable declaration is undefined

Initial value: 5

Value after increment is: 6

Value after addition is: 13

Value after decrement is: 12

The remainder is: 0

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 3</title>
</head>
<body>

  <h2>Variable Operations</h2>

  <script>

    let number;

    document.write("Value after variable declaration is: " + number + "<br><br>");

    number = 5;

    document.write("Initial value: " + number + "<br>");

    number++;

    document.write("Value after increment is: " + number + "<br>");

    number += 7;

    document.write("Value after addition is: " + number + "<br>");

    number--;

    document.write("Value after decrement is: " + number + "<br>");

    document.write("The remainder is: " + (number % 3));

  </script>

</body>
</html>
```

Output:

```
Variable Operations

Value after variable declaration is: undefined

Initial value: 5

Value after increment is: 6

Value after addition is: 13

Value after decrement is: 12

The remainder is: 0
```

4. Cost of one movie ticket is 600 PKR. Write a script to store ticket price in a variable & calculate the cost of buying 5 tickets to a movie.

Example output:



Total cost to buy 5 tickets to a movie is 3000PKR

Code:

```
<html>
<head>
  <title>Question 4</title>
</head>
<body>

  <h2>Movie Ticket Cost</h2>

  <script>

    let ticketPrice = 600;
    let quantity = 5;
    let totalCost = ticketPrice * quantity;

    document.write("Price of one movie ticket is " + ticketPrice + " PKR<br>");
    document.write("Number of tickets = " + quantity + "<br><br>");
    document.write("Total Cost to buy " + quantity + " tickets = " + totalCost + " PKR");

  </script>

</body>
```

Output:

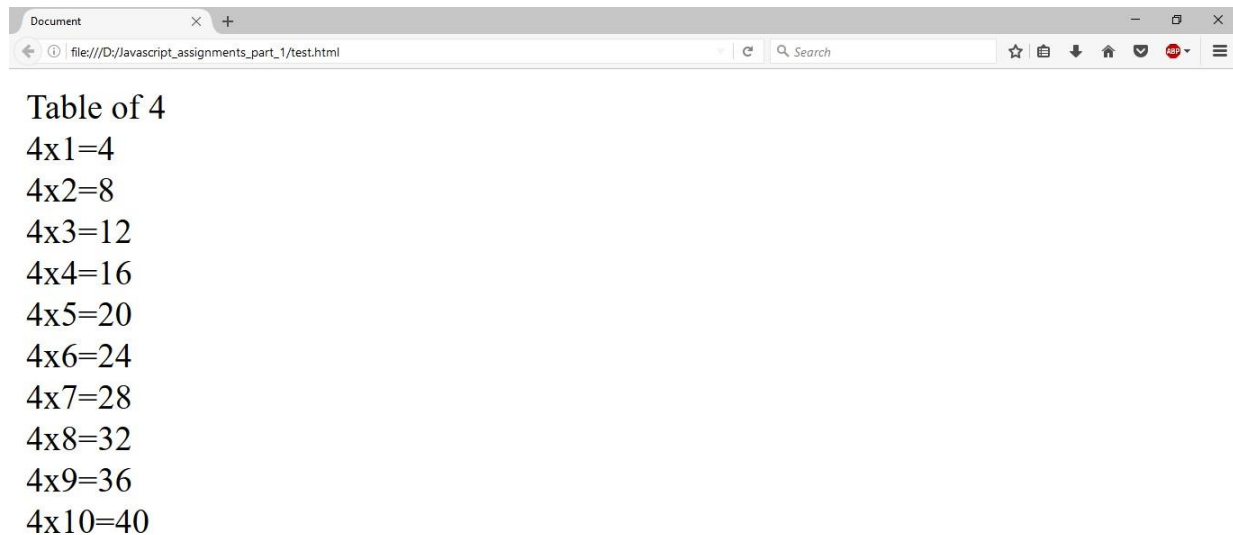
Movie Ticket Cost

Price of one movie ticket is 600 PKR

Number of tickets = 5

Total Cost to buy 5 tickets = 3000 PKR

5. Write a script to display multiplication table of any number in your browser. E.g



CODE:

```
<title>Question 5</title>
</head>
<body>

<h2>Multiplication Table</h2>

<script>

let number = 5;

document.write("<h3>Table of " + number + "</h3>");

for(let i = 1; i <= 10; i++)
{
    document.write(number + " x " + i + " = " + (number * i) + "<br>");
}

</script>

</body>
</html>
```

OUTPUT:

Multiplication Table

Table of 5

$5 \times 1 = 5$
 $5 \times 2 = 10$
 $5 \times 3 = 15$
 $5 \times 4 = 20$
 $5 \times 5 = 25$
 $5 \times 6 = 30$
 $5 \times 7 = 35$
 $5 \times 8 = 40$
 $5 \times 9 = 45$
 $5 \times 10 = 50$

6. The Temperature Converter: It's hot out! Let's make a converter based on the steps here.

- Store a Celsius temperature into a variable.
- Convert it to Fahrenheit & output "NN°C is NN°F".
- Now store a Fahrenheit temperature into a variable.
- Convert it to Celsius & output "NN°F is NN°C".

Conversion Formulae:

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5 / 9$$

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 9 / 5) + 32$$



25°C is 77°F

70°F is 21.11111111111111°C

CODE:

```
<title>Question 6</title>
</head>
<body>

<h2>Temperature Converter</h2>

<script>

let celsius = 25;
let fahrenheit = (celsius * 9 / 5) + 32;

document.write(celsius + "°C is " + fahrenheit + "°F<br><br>");

let fahrenheitTemp = 70;
let celsiusTemp = (fahrenheitTemp - 32) * 5 / 9;

document.write(fahrenheitTemp + "°F is " + celsiusTemp.toFixed(2) + "°C");

</script>

</body>
</html>
```

OUTPUT:

Temperature Converter

25°C is 77°F

78°F is 21.11°C

7. Write a program to implement checkout process of a shopping cart system for an e-commerce website. Store the following in variables

Price of item 1

Price of item 2

Ordered quantity of item 1

Ordered Quantity of item 2

Shipping charges

Compute the total cost & show the receipt in your browser.



Shopping Cart

Price of item 1 is 650
 Quantity of item 1 is 3
 Price of item 2 is 100
 Quantity of item 2 is 7
 Shipping Charges 100

Total cost of your order is 2750

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 7</title>
</head>
<body>

<h2>Shopping Cart</h2>

<script>

let priceItem1 = 650;
let priceItem2 = 100;
let quantityItem1 = 3;
let quantityItem2 = 7;
let shippingCharges = 100;

let totalCost = (priceItem1 * quantityItem1) + (priceItem2 * quantityItem2) + shippingCharges;

document.write("Price of Item 1 is " + priceItem1 + "<br>");
document.write("Quantity of Item 1 is " + quantityItem1 + "<br>");
document.write("Price of Item 2 is " + priceItem2 + "<br>");
document.write("Quantity of Item 2 is " + quantityItem2 + "<br>");
document.write("Shipping Charges " + shippingCharges + "<br><br>");
document.write("<b>Total Cost of your Order is " + totalCost + "</b>");

</script>

</body>
</html>
```

Output:

```
Shopping Cart

Price of Item 1 is 650
Quantity of Item 1 is 3
Price of Item 2 is 100
Quantity of Item 2 is 7
Shipping Charges 100

Total Cost of your Order is 2750
```

8. Store total marks & marks obtained by a student in 2 variables. Compute the percentage & show the result in your browser



Marks Sheet

Total marks: 980
Marks obtained: 804
Percentage: 82.0408163265306%

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 8</title>
</head>
<body>

  <h2>Marks Sheet</h2>

  <script>

    let totalMarks = 980;
    let marksObtained = 804;
    let percentage = (marksObtained / totalMarks) * 100;

    document.write("Total Marks: " + totalMarks + "<br>");
    document.write("Marks Obtained: " + marksObtained + "<br>");
    document.write("Percentage: " + percentage.toFixed(2) + "%");

  </script>

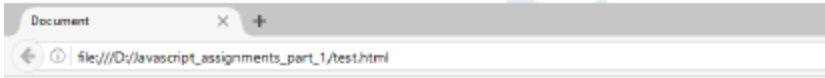
</body>
```

Output:

```
Marks Sheet

Total Marks: 980
Marks Obtained: 804
Percentage: 82.04%
```

9. Assume we have 10 US dollars & 25 Saudi Riyals. Write a script to convert the total currency to Pakistani Rupees. Perform all calculations in a single expression. (Exchange rates : **1 US Dollar = 104.80 Pakistani Rupee** and **1 Saudi Riyal = 28 Pakistani Rupee**)



Currency in PKR

Total Currency in PKR: 1748

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 9</title>
</head>
<body>

<h2>Currency in PKR</h2>

<script>

let totalPKR = (10 * 104.80) + (25 * 28);

document.write("Total Currency in PKR: " + totalPKR);

</script>

</body>
</html>
```

Output:

Currency in PKR

Total Currency in PKR: 1748

10. Write a program to initialize a variable with some number and do arithmetic in following sequence: a. Add 5 b. Multiply by 10 c. Divide the result by 2 Perform all calculations in a single expression.

Code:

```
<html>
<head>
  <title>Question 10</title>
</head>
<body>

<h2>Arithmetic Sequence</h2>

<script>

let number = 8;

let result = ((number + 5) * 10) / 2;

document.write("Initial Number: " + number + "<br>");
document.write("Final Result: " + result);

</script>

</body>
</html>
```

OUTPUT:

```
Arithmetic Sequence

Initial Number: 8
Final Result: 65
```

11. **The Age Calculator:** Forgot how old someone is? Calculate it!

- e. Store the current year in a variable.
- f. Store their birth year in a variable.
- g. Calculate their 2 possible ages based on the stored values.

Output them to the screen like so: “They are either NN or NN years old”.



Age Calculator

Current Year: 2016
Birth Year: 1992
Your Age is: 24

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 11</title>
</head>
<body>

<h2>Age Calculator</h2>

<script>

let currentYear = 2026;
let birthYear = 2004;

let age1 = currentYear - birthYear;
let age2 = age1 - 1;

document.write("Current Year: " + currentYear + "<br>");
document.write("Birth Year: " + birthYear + "<br>");
document.write("They are either " + age2 + " or " + age1 + " years old.");

</script>
```

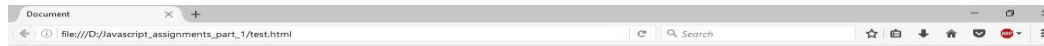
Output:

Age Calculator

Current Year: 2026
Birth Year: 2004
They are either 21 or 22 years old.

12. The Geometrizer: Calculate properties of a circle. a. Store a radius into a variable.

b. Calculate the circumference based on the radius, and output “The circumference is NN”. (*Hint : Circumference of a circle = $2 \pi r$, $\pi = 3.142$*) Calculate the area based on the radius, and output “The area is NN”. (*Hint : Area of a circle = πr^2 , $\pi = 3.142$*)



The Geometrizer

Radius of a circle: 20
 The circumference is: 125.67999999999999
 The area is: 1256.8

CODE:

```
</head>
<body>

<h2>The Geometrizer</h2>

<script>

let radius = 20;
let pi = 3.142;

let circumference = 2 * pi * radius;
let area = pi * radius * radius;

document.write("Radius of a Circle: " + radius + "<br>");
document.write("The Circumference is " + circumference.toFixed(2) + "<br>");
document.write("The Area is " + area.toFixed(2));

</script>

</body>
</html>
```

OUTPUT:

```
The Geometrizer

Radius of a Circle: 20
The Circumference is 125.68
The Area is 1256.80
```


13.The Lifetime Supply Calculator: Ever wonder how much a “lifetime supply” of your favorite snack is? Wonder no more.

- a. Store your favorite snack into a variable
- b. Store your current age into a variable.
- c. Store a maximum age into a variable.
- d. Store an estimated amount per day (as a number).
- e. Calculate how many would you eat total for the rest of your life.

Output the result to the screen like so: “You will need NNNN to last you until the ripe old age of NN”.

Code:

```
<script>

let favoriteSnack = "Chocolate Chip";
let currentAge = 22;
let maximumAge = 80;
let amountPerDay = 3;

let totalNeeded = (maximumAge - currentAge) * 365 * amountPerDay;

document.write("Favourite Snack: " + favoriteSnack + "<br>");
document.write("Current Age: " + currentAge + "<br>");
document.write("Estimated Maximum Age: " + maximumAge + "<br>");
document.write("Amount of Snack per Day: " + amountPerDay + "<br><br>");

document.write("You will need " + totalNeeded + " " + favoriteSnack + " to last you until the ripe old age of " + maximumAge);

</script>

</body>
</html>
```



The Lifetime Supply Calculator

Favourite Snack: chocolate chip

Current age: 15

Estimated Maximum Age: 65

Amount of snacks per day: 3

You will need 150 chocolate chip to last you until the ripe old age of 65

Output:

Lifetime Supply Calculator

Favourite Snack: Chocolate Chip

Current Age: 22

Estimated Maximum Age: 80

Amount of Snack per Day: 3

You will need 63510 Chocolate Chip to last you until the ripe old age of 80.

